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PERFORMANCE SPECIFICATION

VALVES, CHECK, AIRCRAFT LOW PRESSURE OXYGEN SYSTEMS



Comments, suggestions, or questions on this document should be addressed to: Oklahoma City Air Logistics Center/ENSDAA, 3001 Staff Drive, Tinker AFB, OK. 73145-3036 or emailed to af71@tinker.af.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 4820

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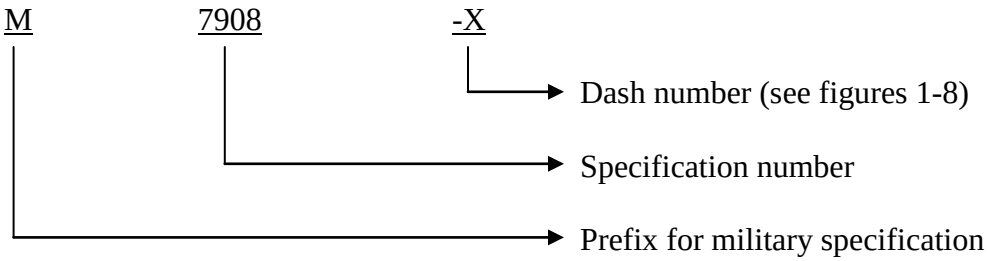
1. SCOPE

1.1 Scope. This specification covers aircraft low pressure oxygen check valves.

1.2 Classification. The valves should be one of the following type and style (see 6.2):

PIN (see 6.9)	TYPE	STYLE
M7908-1 -	Check	flared tube to flared tube
M7908-2 -	Dual check	tee, end pipe thread to flared tube
M7908-3 -	Dual check	tee, side pipe thread to flared tube
M7908-4 -	Dual check	tee, flared tube
M7908-5 -	Triple check	cross, flared tube
M7908-6 -	Check	pipe thread to flared tube
M7908-7 -	Check	flared tube to pipe thread
M7908-8 -	Check	side pipe thread to flared tube

1.3 Part or identifying number (PIN). The PIN to be used for valves acquired to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

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(Copies of these documents are available from http://www.asme.org or ASME Information Central Orders/Inquiries, P.O. Box 2300 Fairfield, NJ 07007-2300).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(ASQC documents may be obtained at http://asq.org or addressed to the American Society for Quality P.O. Box 3005 Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, Wisconsin, 53203, USA)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	-	Salt Spray (Fog) Apparatus, Operating

(ASTM documents may be obtained at http://www.astm.org or addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS8010	-	Aviator's Breathing Oxygen Purity Standard
SAE AS4395	-	Fitting End, Flared, Tube Connection, Design Standard
SAE AS71051	-	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT – Design And Inspection Standard

(SAE documents may be obtained at http://www.sae.org/ or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.3 Materials. All materials shall be corrosion resistant or suitably treated to resist

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corrosion due to electrolytic decomposition, salt fog, and any other atmospheric condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Safety specific materials. All material used shall be capable of operating safely in a pure oxygen environment without creating a health or fire hazard (see 6.3 and 6.7).

3.4 Interface.

3.4.1 Dimensions. The dimensions of the valve shall be in accordance with the applicable valve type and style shown in figures 1 through 8.

3.4.2 Threads. Tapered pipe threads shall be in accordance with SAE AS71051.

3.4.3 Gripping surfaces. There shall be either a hexagonal surface or flats adjacent to each external threaded section. A valve constructed in parts shall have a hexagonal surface or flats on each individual external part. The hexagonal surface and flats shall fit open end wrenches.

3.4.4 Flow markings. The body of the valve shall be permanently and legibly marked with arrow(s) to signify the direction(s) of flow.

3.5 Performance.

3.5.1 Flow direction. The valve shall allow flow in the direction(s) indicated by the arrow(s) on the valve exterior as shown in figures 1 through 8. Flow in the opposite direction(s) shall be checked.

3.5.2 Orientation. The valve shall operate in any position.

3.5.3 Weight. The weight of the valve shall be in accordance with figures 1 through 8.

3.5.4 Leakage.

3.5.4.1 Body leakage. The valve shall not leak while being pressurized from 0 to 450 pounds per square inch gage (psig).

3.5.4.2 Check leakage. The leakage of the valve shall be no greater than 1 cubic centimeter per minute (cm^3/min) when applying 10 to 450 psig in the checked direction.

3.5.5 Pressure drop. The pressure drop through the valve shall not exceed 10 psig.

3.5.6 Proof pressure. The valve shall withstand a pressure of 750 psig.

3.5.7 Endurance. The valve shall withstand 20,000 cycles of opening and closing.

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3.5.8 Valve ignition. The valve shall not ignite or show evidence of charring or burning when subjected to 450 psig with a minute leak (see 6.6).

3.5.9 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 milligrams per square foot (mg/ft^2)(see6.4).

3.6 Environmental conditions. The valve shall operate in the following conditions:

a. Temperature: -65° to 160°F.
b. Vibration: Frequencies from 5 to 50 Hz at a double amplitude of 0.018 and 0.020 inch.

3.7 Item identification. The valve shall be marked with the following information (see 6.2):

a. Item name.
b. Manufacturer's identification.
c. Manufacturer's CAGE code.
d. Manufacturer's part number (PN).
e. Part or Identification Number (PIN).
f. Additional identification information (see 6.2).

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (see 4.2).
b. Conformance (see 4.3).

4.2 First article inspection. First article inspection shall be performed on three valves of each type. First article inspection shall include all tests described in 4.6.

4.3 Conformance inspection. Conformance inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual tests. Each valve shall be subjected to the following tests:

a. Examination (see 4.6.1).
b. Check leakage (see 4.6.4.2).
c. Pressure drop (see 4.6.5).

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4.3.2 Sample tests. In addition to the tests indicated in 4.3.1, the valves and fittings shall be sampled in accordance with ANSI/ASQ Z1.4 at a normal inspection level unless modified in the procurement document (see 6.2), and shall be subjected to the following tests:

a. Dimensions (see 4.6.2).
b. Body leakage (see 4.6.4.1).
c. Proof pressure (see 4.6.8).

4.4 Test conditions.

4.4.1 Test medium. Oxygen conforming to SAE AS 8010, Type I, shall be used.

4.4.2 Temperature and pressure. Unless otherwise specified in the individual test description, tests shall be conducted at ambient temperature and pressure.

4.4.3 Multiple checks. For those valve types that have more than one check devices, tests performed shall be applicable to each flow passage through the valve.

4.4.4 Positioning. Valves subjected to leakage and pressure drop tests shall be positioned with the center line of the passage parallel to a vertical plane. Each consecutive valve tested shall be rotated 90° about an axis perpendicular to the vertical plane from the position of the valve previously tested. The rotation shall always be in the same direction.

4.5 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE I. Requirements cross-reference matrix

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.4.1	4.6.4.1
3.3	4.6.1, 4.6.11, 4.6.12	3.5.4.2	4.6.4.2
3.3.1	4.6.1, 4.6.10, 4.6.12	3.5.5	4.6.5
3.4.1	4.6.2	3.5.6	4.6.8
3.4.2	4.6.1	3.5.7	4.6.9
3.4.3	4.6.1	3.5.8	4.6.10
3.4.4	4.6.1	3.5.9	4.6.3
3.5.1	4.6.4.2	3.6	4.6.6, 4.6.7
3.5.2	4.6.4	3.7	4.6.1
3.5.3	4.6.1	3.8	4.6.1

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4.6 Tests.

4.6.1 Examination. The valve shall be examined for compliance with the requirements for materials, interface (less dimensions), weight, interchangeability, markings, and item identification (see 6.3).

4.6.2 Dimensions. The valve shall be examined for compliance with the requirements for dimensions.

4.6.3 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 mg/ft² (see 6.4). Cleanliness of the surfaces shall be demonstrated by industrially accepted methods and these cleaning and verification methods shall be identified (see 6.2 and 6.4).

4.6.4 Leakage.

4.6.4.1 Body leakage. With the outlet sealed, the valve shall be immersed in water and the inlet(s) shall be pressurized from 0 to 450 psig. While the valve is immersed, a pressure of 450 psig shall be maintained for 2 minutes. There shall be no breakaway of bubbles after 30 seconds (see 4.4.4 and 6.5).

4.6.4.2 Check leakage. The outlet of the valve shall be pressurized from 0 to 450 psig at a rate between 200 and 450 psig/minute. The inlet(s) leakage shall not exceed one cm^3/min (see 4.4.4).

4.6.5 Pressure drop. A pressure of 300 psig shall be applied at the inlet(s) of the valve and a flow of 400 standard liters per minute (SLPM)(see 6.8) shall be maintained through the valve for 1 minute. The pressure drop across the valve shall not exceed 10 psig (see 4.4.4).

4.6.6 High and low temperature.

4.6.6.1 High temperature. With a pressure of 450 psig maintained at the outlet, the valve shall be subjected to a temperature of 160° ± 5°F for 3 hours. At the beginning, in the middle, and at the end of this period, the valve shall be subjected to the check leakage test in 4.6.4.2

4.6.6.2 Low temperature. With a pressure of 450 psig maintained at the outlet, the valve shall be subjected to a temperature of -65° ± 5°F for 3 hours. At the beginning, in the middle, and at the end of period, the valve shall be subjected to the check leakage test in 4.6.4.2

4.6.7 Vibration. A pressure of 450 psig shall be applied to the outlet of the valve. The valve shall be vibrated for 6 hours using a double amplitude of 0.018 to 0.020 inch at a varying frequency of 5 to 50 Hz. During the vibration period, the valve shall be subjected to the check leakage and pressure drop tests in 4.6.4.2 and 4.6.5 at six equal intervals

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4.6.8 Proof pressure. A pressure of 750 psig shall be applied to the inlet(s) with the outlet sealed and maintained for 1 minute. The valve shall then be subjected to the body leakage and check leakage tests in 4.6.4.1 and 4.6.4.2.

4.6.9 Endurance. With an inlet pressure of 450 psig, a flow of 50 SLPM shall be induced through the valve for 3 seconds (opening the valve) followed by a 0 SLPM flow for 3 seconds (closing the valve). This shall constitute one cycle. The test shall continue for a minimum of 20,000 cycles. The valve shall then be subjected to the individual tests in 4.3.1 during the test, the pressure 450 psig may drop to 300 psig before recharging to 450 psig.

4.6.10 Valve ignition. With a pressure of 450 psig applied to the valve outlet, the valve shall be opened to create a minute leak (see 6.6). This test shall be conducted three times for a period of 5 minutes each. The valve shall show no evidence of charring or burning

4.6.11 Salt fog. With the inlet(s) and outlet plugged, the valve shall be subjected to the salt fog test in ASTM B117 for 50 hours. There shall be no damage and the valve shall be subjected to the leakage tests in 4.6.4

4.6.12 Disassembly and examination. At the conclusion of these tests, the valve shall be disassembled and show no evidence of deterioration or wear.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The check valves are intended for use in low pressure, 450 psig or less, aircraft oxygen systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Classification (see 1.2)
c. If first article is required (see 3.1).
d. Additional item identification required, if any (see 3.7).

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e. If the inspection level for sampling is different than normal (i.e., tightened or reduced)(see 4.3)
f. The requirement for the vendor to identify proposed cleaning and verification methods (see 4.6.3).
g. Packaging requirements (see 5.1).

6.3 Materials. Material certification has been used to demonstrate compliance with the safety specific and corrosion requirement.

6.4 Cleaning. See MIL-STD-1330 and MIL-STD-1359 for guidance on proven cleaning methods and verifications. Visible inspections are typically conducted using white light and ultraviolet light to detect particulates and some types of hydrocarbons. The non-volatile residue (NVR) test in MIL-STD-1359 may be used to baseline the hydrocarbon verification at an acceptable contamination level, however, other cleaning and verification methods that do not contain class I and II ODC solvents should be used for production.

6.5 Breakaway of bubbles. The breakaway of bubbles is the formation and breaking away of bubbles on the surface of a submerged test valve while applying internal pressure. The formation of bubbles is acceptable but breaking away is not.

6.6 Minute leak. A minute leak occurs at the point where the valve just begins to open and creates a leak with a constant flow. This leak creates a high-speed concentrated oxygen flow across the sealing surfaces of the valve, which causes them to superheat.

6.7 Health hazards. Materials such as cadmium, lead, and polyvinyl chloride can be susceptible to outgasing in the presence of pure oxygen and at elevated temperatures. Outgasing by these and many other materials when used in breathing oxygen equipment can cause potential health hazards.

6.8 SLPM. One SLPM is equal to one liter of flow per one minute at a pressure of 760 millimeters of mercury at the temperature of 0° Celsius.

6.9 Supersession data. For supersession data see table II.

TABLE II. Supersession data

New PIN	Old part number
M7908-1	MS21211-1
M7908-2	MS21211-2
M7908-3	MS21211-3
M7908-4	MS21211-4
M7908-5	MS21211-5
M7908-6	MS21211-6
M7908-7	MS21211-7
M7908-8	MS21211-8

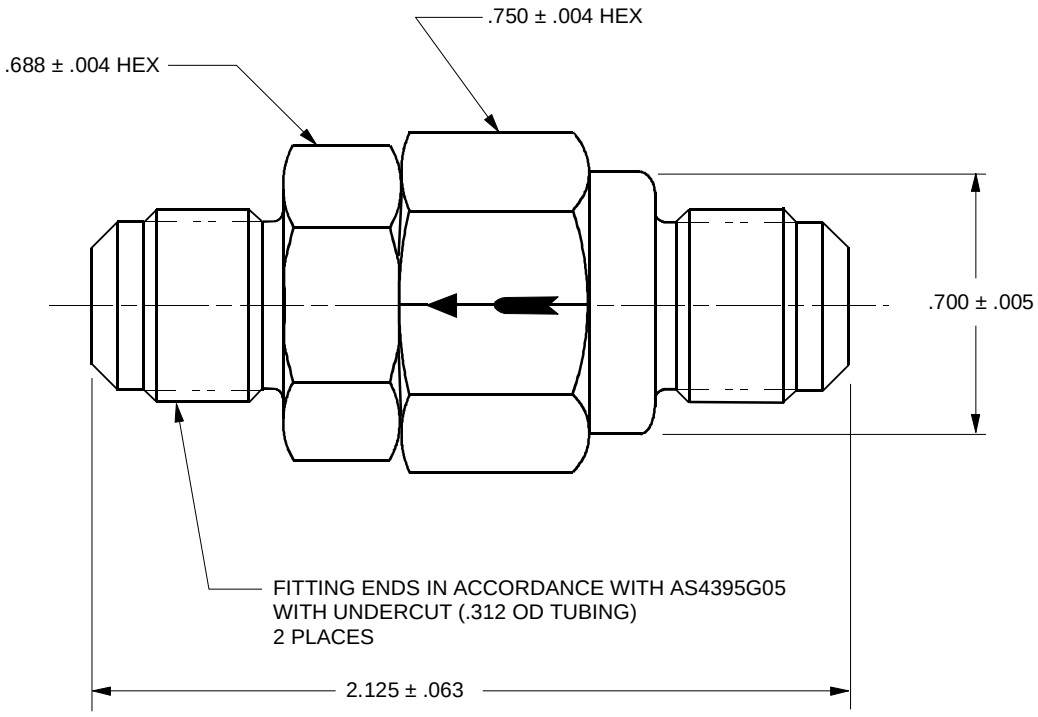
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6.10 Subject term (key word) listing.

Breathing
450 psig

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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PIN	MAX WEIGHT	SUPERSEDING
M7908-1	.07 POUND	MS21211-1

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

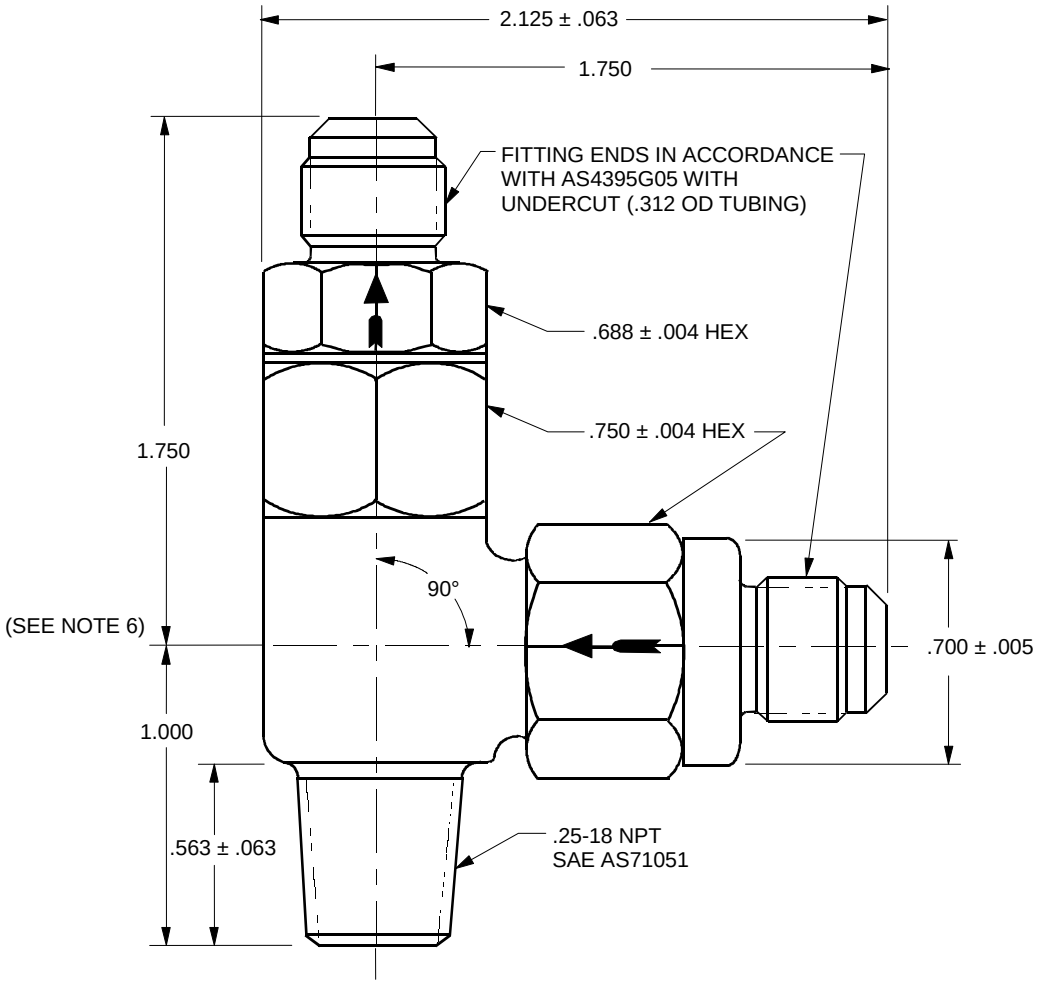
3. DIMENSIONS IN INCHES.

4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

FIGURE 1. Valve, check, flared tube to flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-2	.14 POUND	MS21211-2

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

3. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031; ANGLES ±1°.

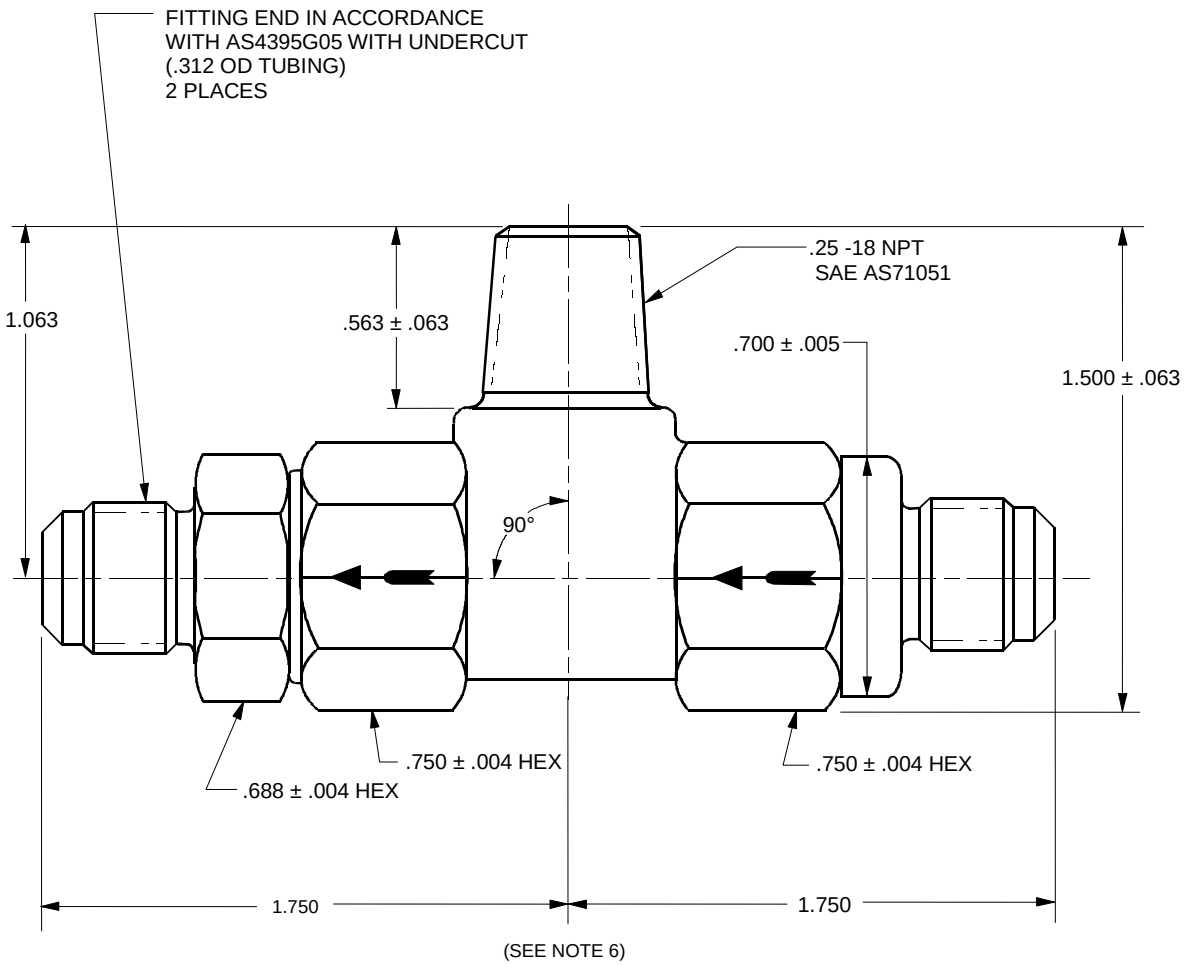
4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

6. TOLERANCE FOR OVERALL LENGTH ±.063.

FIGURE 2. Valve, dual check, tee, end pipe thread to flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-3	.15 POUND	MS21211-3

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

3. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031; ANGLES ±1°.

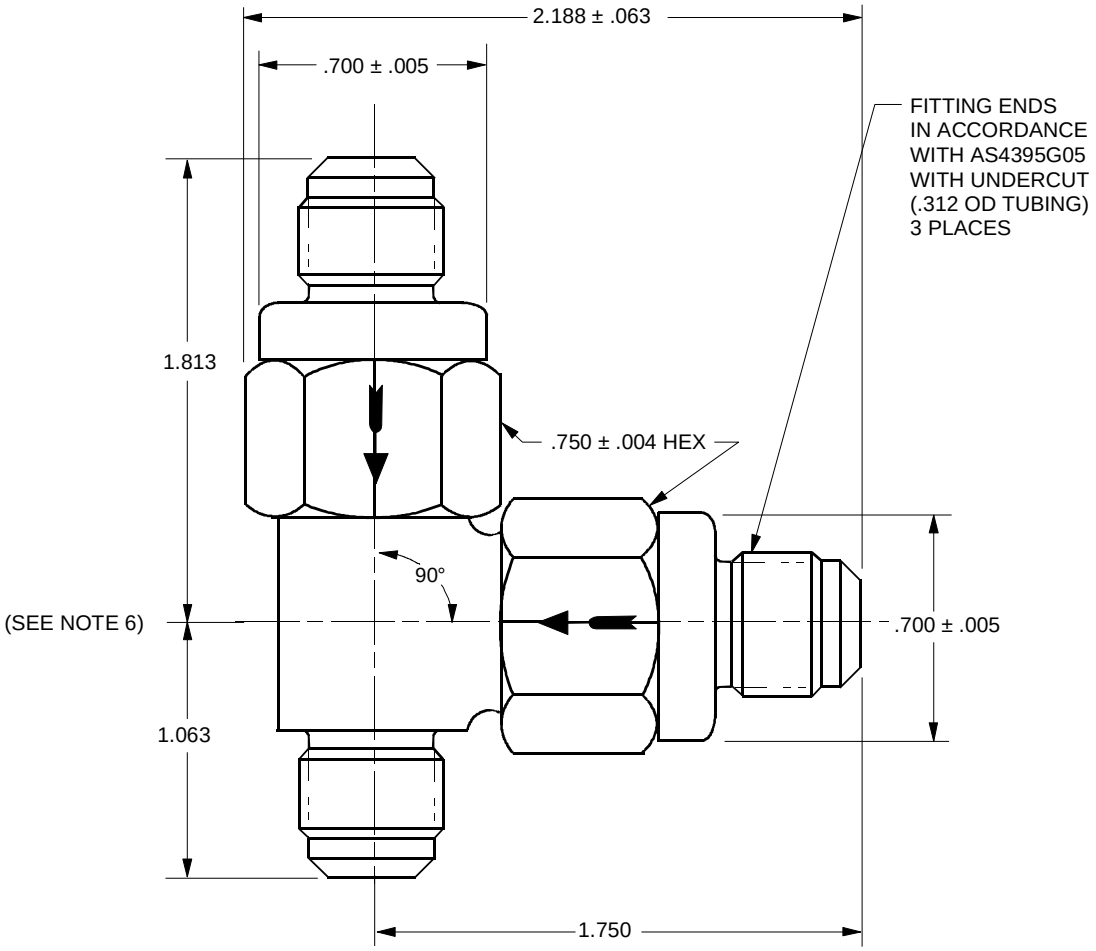
4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

6. TOLERANCE FOR OVERALL LENGTH ±.063.

FIGURE 3. Valve, dual check, tee, side pipe thread to flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-4	.155 POUND	MS21211-4

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

3. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031; ANGLES ±1°.

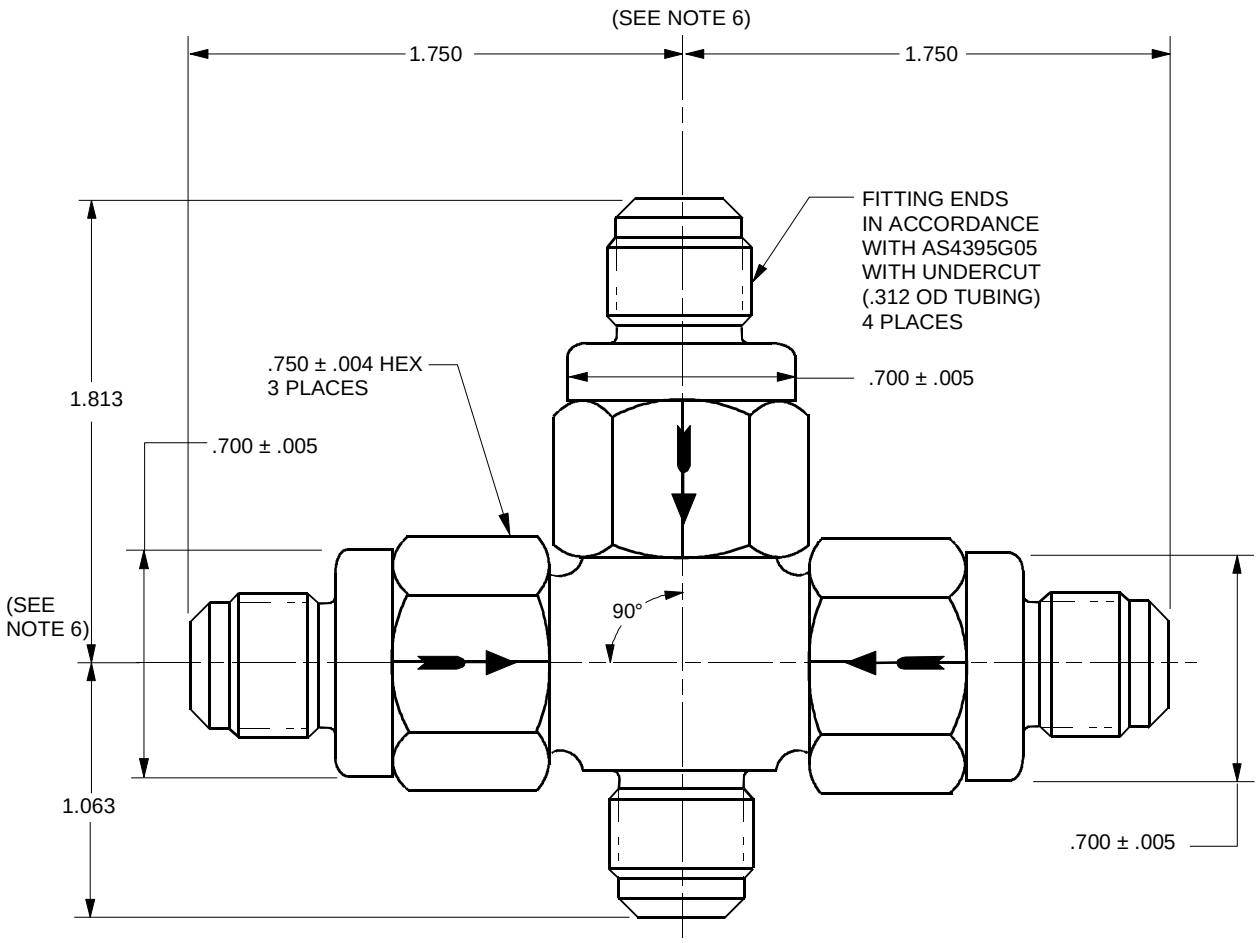
4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

6. TOLERANCE FOR OVERALL LENGTH ±.063.

FIGURE 4. Valve, dual check, tee, flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-5	.20 POUND	MS21211-5

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

3. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031; ANGLES ±1°.

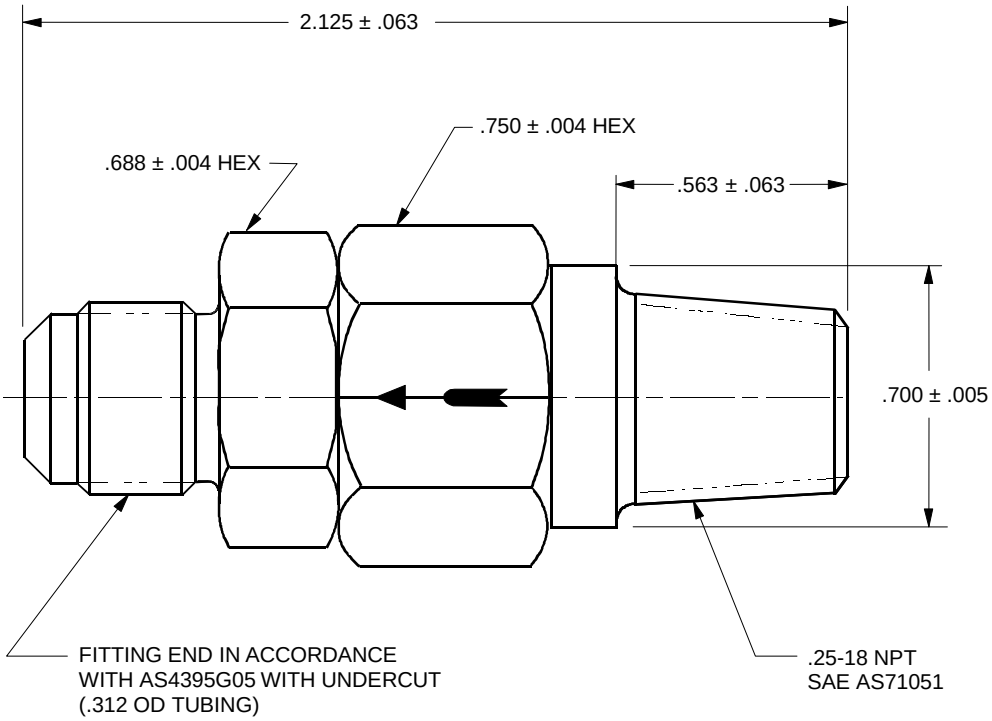
4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

6. TOLERANCE FOR OVERALL LENGTH ±.063.

FIGURE 5. Valve, triple check, cross, flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-6	.07 POUND	MS21211-6

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

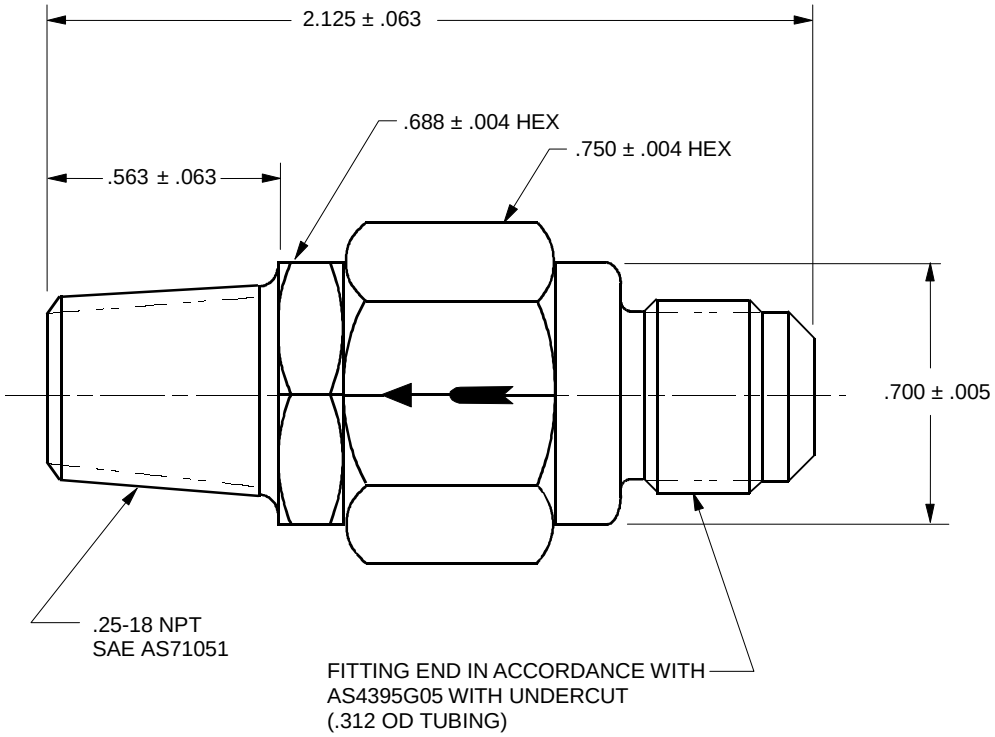
3. DIMENSIONS IN INCHES.

4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

FIGURE 6. Valve, check, pipe thread to flared tube

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PIN	MAX WEIGHT	SUPERSEDING
M7908-7	.07 POUND	MS21211-7

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

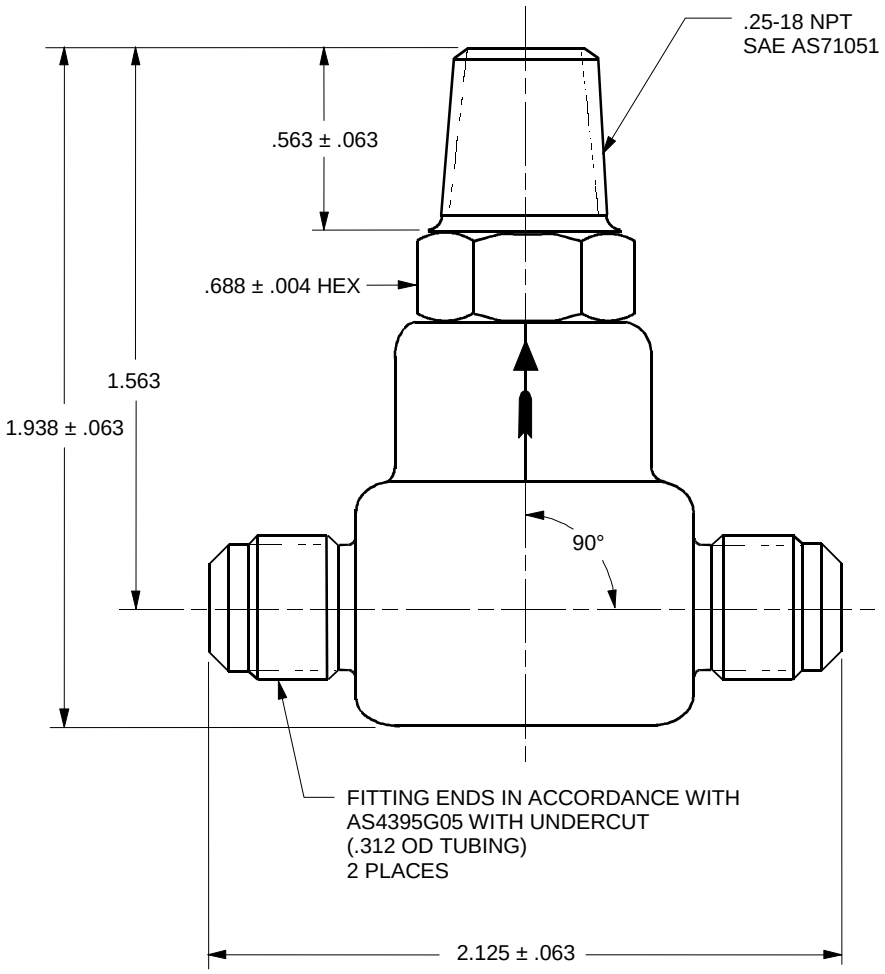
3. DIMENSIONS IN INCHES.

4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

FIGURE 7. Valve, check, flared tube to pipe thread

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PIN	MAX WEIGHT	SUPERSEDING
M7908-8	.11 POUND	MS21211-8

NOTES: 1. UNLESS OTHERWISE SPECIFIED, MAXIMUM ROUGHNESS SHALL BE 125 IN ACCORDANCE WITH ASME B46.1.

2. OPPOSITE FITTING ENDS SHALL BE CONCENTRIC WITH EACH OTHER WITHIN .032 FULL INDICATOR READING.

3. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031; ANGLE ±1°.

4. THE SUPERSEDING DASH NUMBER IS DIMENSIONALLY AND FUNCTIONALLY INTERCHANGEABLE WITH THE SUPERSEDED MS NUMBER.

5. FIGURE NOT TO SCALE.

FIGURE 8. Valve, check, side pipe thread to flared tube

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Custodians:
Air Force - 71
Army - AV
Navy - AS

Preparing activity:
Air Force - 71

(Project 4820-2009-023)

Review activities:
Air Force - 11
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/

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